

SURVEY QUESTIONNAIRE

Design of an AI-Assisted Customer Support System for Micro and Small E-Commerce Businesses in Baguio City and Benguet

Dear Respondent,

Magandang araw po! I am Jeanalyn Marie B. Lagera, a graduate student at Mapúa University. For my research project, I am designing a customer support assistant for small online sellers in Baguio City and Benguet. The assistant would answer common customer questions (stock, price, payment, delivery, returns) using information that the seller approves, and it would pass complaints and difficult questions to a real person.

Before I build it, I want to learn how sellers like you handle customer messages today and what you would actually find useful. This survey takes about 10 to 15 minutes. There are no right or wrong answers.

Your participation is voluntary, and you may skip any question or stop at any time. Please do not write your customers' names, phone numbers, or addresses anywhere in this form. Your answers will be kept confidential, reported only in summary form, and used only for this study, in line with the Data Privacy Act of 2012 (RA 10173). The data will be deleted after the study is completed.

Maraming salamat po for your time and help.

Jeanalyn Marie B. Lagera

Researcher | jeanalynlagera123@gmail.com

INFORMED CONSENT

☐ I have read the information above. I understand that my participation is voluntary and that my answers will be kept confidential. I agree to take part in this survey.

Signature over printed name (optional)

Date

Who should answer: owners or staff of micro and small businesses in Baguio City or Benguet that sell products online (Facebook, Marketplace, Instagram, TikTok, Shopee, Lazada, or their own website) and reply to customer messages.

SURVEY A. SELLER NEEDS ASSESSMENT

To be answered before the prototype is built. Please check (✓) the box that applies or write your answer on the line.

Screening Question

S1. Do you sell products, or receive customer orders or inquiries, through any online channel (Facebook page, Messenger, Facebook Marketplace, Instagram, TikTok, Shopee, Lazada, Viber, or a website)?

- ☐ Yes. Please continue to Part I.
☐ No. Thank you for your time. You may stop here. (Optional) Main reason for not selling online:

Part I. Business Profile

1. Where is your business located?

- | | | |
|--------------------------------------|---------------------------------|--|
| ☐ Baguio City | ☐ Itogon | ☐ Sablan |
| ☐ La Trinidad | ☐ Tuba | ☐ Others: _____ |

2. What is your role in the business?

- | | |
|--|--|
| ☐ Owner | ☐ Family member helping out |
| ☐ Owner and main person replying to customers | ☐ Others: _____ |
| ☐ Staff / customer service | |

3. What do you mainly sell?

- | | |
|--|--|
| ☐ Food, pasalubong, or delicacies | ☐ Home goods |
| ☐ Coffee or farm products | ☐ Gadgets and accessories |
| ☐ Clothing / ukay-ukay | ☐ Plants and flowers |
| ☐ Handicrafts or woven products | ☐ Others: _____ |
| ☐ Beauty and personal care | |

4. How long has your business been selling online?

- | | |
|---|--|
| ☐ Less than 1 year | ☐ 3 to 5 years |
| ☐ 1 to 2 years | ☐ More than 5 years |

5. How many people (including you) work in the business?

- | | |
|----------------------------------|-------------------------------------|
| ☐ Just me | ☐ 4 to 9 |
| ☐ 2 to 3 | ☐ 10 or more |

6. How many people reply to customer messages?

- | | |
|----------------------------------|------------------------------------|
| ☐ Only me | ☐ 3 or more |
| ☐ 2 | |

7. Where do you sell online? (Check all that apply.)

- | | |
|---|---|
| ☐ Facebook page | ☐ Lazada |
| ☐ Facebook Marketplace | ☐ Own website |
| ☐ Instagram | ☐ Viber or WhatsApp groups |
| ☐ TikTok / TikTok Shop | ☐ Others: _____ |
| ☐ Shopee | |

8. Have you joined any digital selling or e-commerce training? (Check all that apply.)

- | | |
|---|---|
| ☐ DTI program or Negosyo Center | ☐ Tatak Pinoy roadshow |
| ☐ DICT SPARK / digital skills training | ☐ None yet |
| ☐ Shopee / Lazada seller training | ☐ Others: _____ |

Part II. Current Customer Support Practices

9. On a normal day, about how many customer messages or inquiries do you receive?

- | | | |
|-----------------------------------|------------------------------------|--|
| ☐ 1 to 10 | ☐ 31 to 50 | ☐ More than 100 |
| ☐ 11 to 30 | ☐ 51 to 100 | |

10. On busy days (sale days, holidays, Panagbenga season), about how many do you receive?

- | | | |
|-----------------------------------|------------------------------------|--|
| ☐ 1 to 10 | ☐ 31 to 50 | ☐ More than 100 |
| ☐ 11 to 30 | ☐ 51 to 100 | |

11. Where do most customer questions come from? (Check up to two.)

- | | |
|---|--|
| ☐ Messenger / page inbox | ☐ TikTok messages |
| ☐ Comments on posts | ☐ Text or phone calls |
| ☐ Marketplace chat | ☐ Others: _____ |
| ☐ Shopee / Lazada chat | |

12. How do you usually reply to customers? (Check all that apply.)

- | | |
|---|--|
| ☐ I type each reply myself | ☐ I use a chatbot tool |
| ☐ I use saved or quick replies | ☐ Someone else replies for me |
| ☐ I use Meta's automated responses / FAQ | ☐ Others: _____ |

13. How long does it usually take you to reply to a customer message?

- | | | |
|---|---|---|
| ☐ Within 5 minutes | ☐ 30 minutes to 1 hour | ☐ More than 3 hours |
| ☐ 5 to 30 minutes | ☐ 1 to 3 hours | ☐ Usually the next day |

14. About how many hours a day do you spend answering customer messages?

- | | |
|---|--|
| ☐ Less than 1 hour | ☐ 3 to 4 hours |
| ☐ 1 to 2 hours | ☐ More than 4 hours |

15. Do you receive messages outside your store hours (late at night or early morning)?

- ☐ Often ☐ Sometimes ☐ Rarely ☐ Never

How often do customers ask about the following? Check one box per row.

Type of question	Very often	Often	Sometimes	Rarely	Never
Is the item available / in stock?	☐	☐	☐	☐	☐
How much is the item?	☐	☐	☐	☐	☐
Sizes, colors, or product details	☐	☐	☐	☐	☐
Payment methods (COD, GCash, bank)	☐	☐	☐	☐	☐
Delivery area, fee, and time	☐	☐	☐	☐	☐
Order status / tracking	☐	☐	☐	☐	☐
Returns, exchanges, or refunds	☐	☐	☐	☐	☐
Store location and hours	☐	☐	☐	☐	☐
Discounts or promos	☐	☐	☐	☐	☐
Complaints (late, wrong, or damaged item)	☐	☐	☐	☐	☐

16. In what language do most of your customers message you?

- ☐ English ☐ Ilocano or other local language
☐ Filipino / Tagalog ☐ Others: _____
☐ Taglish (mixed English and Filipino)

How much do you agree with these statements about your current way of replying?

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
B1	I answer the same questions many times a day.	☐	☐	☐	☐	☐
B2	Replying to messages takes time away from packing, restocking, or other work.	☐	☐	☐	☐	☐
B3	I sometimes miss or forget to reply to a message.	☐	☐	☐	☐	☐
B4	Customers sometimes buy from another seller because I replied late.	☐	☐	☐	☐	☐
B5	My replies are sometimes inconsistent (e.g., different prices or delivery times given to different customers).	☐	☐	☐	☐	☐
B6	It is hard to keep track of changes in stock, prices, or policies when replying.	☐	☐	☐	☐	☐
B7	Complaints and refund requests need my personal attention.	☐	☐	☐	☐	☐

Part III. Views on an AI Customer Support Assistant

17. Have you used an AI tool (e.g., ChatGPT, Meta AI, Gemini) for your business before?

- ☐ Yes, often ☐ No, but I know what it is
☐ Yes, a few times ☐ No, and I am not familiar with it

How much do you agree with the following?

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
C1	An assistant that answers routine questions for me would save me time.	☐	☐	☐	☐	☐
C2	I would trust the assistant if it answered only from the information I approved.	☐	☐	☐	☐	☐
C3	I want to see which product or policy record the assistant used for each answer.	☐	☐	☐	☐	☐
C4	The assistant should reply in the same language as the customer (English, Filipino, or Taglish).	☐	☐	☐	☐	☐
C5	Complaints, refunds, and unusual requests should always be passed to me or my staff.	☐	☐	☐	☐	☐
C6	Customers should be told when they are talking to an automated assistant.	☐	☐	☐	☐	☐
C7	I would be able to update my products, prices, and policies in a simple dashboard.	☐	☐	☐	☐	☐
C8	I am worried the assistant might give wrong information to my customers.	☐	☐	☐	☐	☐
C9	I am worried about the privacy of my customers' information.	☐	☐	☐	☐	☐
C10	I would recommend this kind of tool to other small sellers.	☐	☐	☐	☐	☐

How important is each feature to you? Check one box per row.

Feature	Very important	Important	Somewhat	Not important
Answers stock, price, payment, and delivery questions automatically	☐	☐	☐	☐
Works in English, Filipino, and Taglish	☐	☐	☐	☐
"Talk to staff" button that passes the chat to me	☐	☐	☐	☐
Replies to customers even outside store hours	☐	☐	☐	☐
Simple dashboard for updating products and FAQs	☐	☐	☐	☐
Reminder when product or stock information is outdated	☐	☐	☐	☐
Reports showing the most common customer questions	☐	☐	☐	☐
Connects to Facebook Messenger	☐	☐	☐	☐
Connects to Shopee / Lazada / TikTok Shop chat	☐	☐	☐	☐
Masks card numbers and other sensitive data	☐	☐	☐	☐

18. Which types of questions would you NOT want an AI assistant to answer on its own? (Check all that apply.)

- | | |
|--|--|
| ☐ Refund requests | ☐ Payment problems |
| ☐ Complaints about damaged or wrong items | ☐ Questions about delivery delays |
| ☐ Discount or price negotiation | ☐ Others: _____ |
| ☐ Custom or bulk orders | |

Part IV. Cost and Willingness to Use

19. Would you try the assistant if a free version were available (up to 100 answered conversations a month)?

- ☐ Yes ☐ Maybe ☐ No

20. How much would you be willing to pay per month for this kind of assistant?

- | | | |
|---------------------------------------|---------------------------------------|---|
| ☐ Only if free | ☐ ₱300 to ₱499 | ☐ ₱1,000 to ₱1,499 |
| ☐ Below ₱300 | ☐ ₱500 to ₱999 | ☐ ₱1,500 or more |

21. Would you want someone to help you set up your products and FAQs the first time?

- ☐ Yes, even if there is a one-time fee ☐ No, I can do it myself
☐ Yes, but only if free

22. Which device would you mostly use to manage the dashboard?

- | | |
|--|--|
| ☐ Smartphone | ☐ Tablet |
| ☐ Laptop or desktop | ☐ Both phone and computer |

23. How is your internet connection where you usually work?

- ☐ Stable most of the time ☐ Sometimes slow or unstable ☐ Often slow or unstable

Part V. Open-Ended Questions

24. What are the three questions customers ask you most often? (Please write them the way customers usually type them, e.g., "hm po?", "avail pa?")

25. What is the hardest or most time-consuming part of replying to customers?

26. What would make you hesitate to use an AI assistant for your business?

27. Any other suggestions or features you would like to see?

28. Would you be willing to join a short follow-up interview or test the prototype later?

- ☐ Yes (contact number or email): _____
- ☐ No

29. Do you know other online sellers in Baguio City or Benguet who might be willing to answer this survey? (Optional: page name or contact, shared only with their permission)

Thank you very much for answering this survey!

SURVEY B. PROTOTYPE EVALUATION (POST-TEST)

To be answered after trying the prototype. Respondents will first complete a few short tasks, such as asking about a product's price, asking about COD, adding a new FAQ in the dashboard, and sending a complaint to see how it is passed to staff. The test store and products are fictional.

Part I. Respondent Information

1. I tested the prototype as a:

☐ Seller / business owner or staff

☐ Customer / online shopper

2. Age group:

☐ 18 to 24

☐ 35 to 44

☐ 55 and above

☐ 25 to 34

☐ 45 to 54

3. How often do you buy or sell online?

☐ Daily

☐ Weekly

☐ Monthly

☐ Rarely

4. Which language did you mainly use when chatting with the assistant?

☐ English

☐ Filipino

☐ Taglish

Part II. Accuracy and Relevance of Answers

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
A1	The assistant's answers were correct based on the store information.	☐	☐	☐	☐	☐
A2	The assistant understood what I was asking.	☐	☐	☐	☐	☐
A3	The answers were relevant to my question.	☐	☐	☐	☐	☐
A4	The assistant did not make up information that was not in the store records.	☐	☐	☐	☐	☐
A5	The answers were clear and specific (e.g., gave the actual price or delivery time).	☐	☐	☐	☐	☐
A6	The assistant understood my message even when I used Taglish or informal spelling.	☐	☐	☐	☐	☐

Part III. Response Time

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
B1	The assistant replied quickly.	☐	☐	☐	☐	☐
B2	I did not have to wait long to get the information I needed.	☐	☐	☐	☐	☐
B3	The response time was acceptable for a chat conversation.	☐	☐	☐	☐	☐

Part IV. Usability (System Usability Scale)

These ten items follow the standard System Usability Scale (SUS). Odd-numbered items are positive and even-numbered items are negative; this is intentional.

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
C1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
C2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
C3	I thought the system was easy to use.	☐	☐	☐	☐	☐
C4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
C5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
C6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
C7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
C8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
C9	I felt very confident using the system.	☐	☐	☐	☐	☐
C10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Part V. Trust, Transparency, and Human Escalation

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
D1	It was clear that I was talking to an automated assistant.	☐	☐	☐	☐	☐
D2	I could easily ask to talk to a real person.	☐	☐	☐	☐	☐
D3	When the assistant could not answer, it told me a staff member would reply instead of guessing.	☐	☐	☐	☐	☐
D4	My complaint or refund request was passed to a person and not handled by the assistant alone.	☐	☐	☐	☐	☐
D5	I felt my personal information was handled safely.	☐	☐	☐	☐	☐
D6	I would trust the answers of this assistant when buying from a small seller.	☐	☐	☐	☐	☐

Part VI. For Sellers Only: Owner Dashboard

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
E1	It was easy to add or update a product, price, or stock status.	☐	☐	☐	☐	☐
E2	It was easy to add or edit an FAQ or store policy.	☐	☐	☐	☐	☐
E3	The escalation queue made it easy to see and answer chats that needed me.	☐	☐	☐	☐	☐
E4	The reports helped me understand what customers ask most.	☐	☐	☐	☐	☐
E5	I could manage the system without help from a technical person.	☐	☐	☐	☐	☐
E6	This system would reduce the time I spend answering repeated questions.	☐	☐	☐	☐	☐

Part VII. Overall Satisfaction

Scale: 1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (SA)

No.	Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)
F1	Overall, I am satisfied with the prototype.	☐	☐	☐	☐	☐
F2	The assistant made it easier to get basic store information.	☐	☐	☐	☐	☐
F3	I would use this kind of assistant for routine questions.	☐	☐	☐	☐	☐
F4	I would recommend this assistant to small online sellers.	☐	☐	☐	☐	☐

Part VIII. Comments

5. What did you like most about the assistant?

6. What problems did you encounter, or what did the assistant get wrong?

7. What should be improved before it is used by real businesses?

Thank you for testing the prototype!

RESEARCHER'S GUIDE: SCORING AND ALIGNMENT WITH THE STUDY

This page is for the researcher and adviser only. Remove it before printing copies for respondents.

Rating scale and interpretation

Mean range	Verbal interpretation
4.21 – 5.00	Strongly Agree / Very High
3.41 – 4.20	Agree / High
2.61 – 3.40	Neutral / Moderate
1.81 – 2.60	Disagree / Low
1.00 – 1.80	Strongly Disagree / Very Low

Sampling and screening

- Sampling is purposive: only businesses that sell or take inquiries through an online channel are included. Referral (snowball) sampling is used through the last question of Survey A.
- Respondents who answer "No" to S1 are excluded from the analysis. Count them and report the reasons given, but do not include them in the Survey A percentages or means.
- In Google Forms, set S1 to "Go to section based on answer" so that a "No" answer skips to the end.
- Report the recruitment channels used (DTI/Negosyo Center, DICT-CAR, city office, Facebook groups, bazaars, referrals) and how many responses came from each.

SUS scoring (Survey B, Part IV)

- For odd items (C1, C3, C5, C7, C9): score = response – 1.
- For even items (C2, C4, C6, C8, C10): score = 5 – response.
- Add the ten scores and multiply by 2.5 to get a SUS score from 0 to 100.
- Compare the average SUS score with the project target of 70 or higher (Table 1 of the paper).

Alignment of survey parts with the research

Survey part	Research question / objective addressed	Measure in the paper
A-I, A-II (items and grid)	RQ1: seller needs; Objective 1: requirements	Requirements (Sec. 4.2–4.3)
A-II language item, A-V item 1	RQ3: English, Filipino, Taglish handling	Test-set design (Sec. 3.4)
A-III (C-items, features)	RQ2, RQ4: knowledge base and escalation needs	Business rules (Sec. 4.8)
A-IV	Business model and pricing	Tiers (Sec. 4.13)
B-II	RQ5: accuracy and relevance	Accuracy $\geq 90\%$, relevance ≥ 4.0
B-III	RQ5: response time	Median ≤ 5 s (with system logs)
B-IV	RQ5: usability	SUS ≥ 70
B-V	RQ4: escalation, trust, privacy	Escalation safety 100%
B-VI	Objective 2: owner-managed knowledge base	Maintainability requirement
B-VII	RQ5: satisfaction	Satisfaction ≥ 4.0

Note: Survey B measures user perception. Actual accuracy and response time must also be measured objectively with the 120-item test set and system logs (Appendix A of the paper), so that a high satisfaction score cannot hide factual errors.